FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility

FSSAI: 10012022000345 | ISO 22000:2018 / ISO 14001:2015

EQUIPMENT MANUAL

Siemens CP102 Control Panel - Operation and Maintenance Manual

Model: FF-CP-244 | Serial: FFB-CP-26196

Equipment ID: CP102

Document Number: MAN-CP102-001
Revision: Rev 2.1
Classification: CONFIDENTIAL - INTERNAL USE ONLY
Prepared by: Engineering / QA Department
Approved by: Plant Manager

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1. Introduction and Equipment Description

The CP102 Control Panel (Asset ID: CP102) is a PLC Control Panel manufactured by Siemens, model FF-CP-244, serial number FFB-CP-26196. It is installed in the Utilities department at FreshFlow Bottling Plant - Pune Production Facility. The equipment was commissioned on 2020-05-15 with a warranty valid until N/A.

The CP102 Control Panel is classified as High criticality equipment with an expected operational life of 15 years. It requires weekly preventive maintenance as per the FreshFlow Beverages Asset Management Program (AMP). The equipment is located at: Utilities Zone 3.

All maintenance and operational activities must comply with FreshFlow Beverages Pvt. Ltd. safety and food hygiene standards, ISO 22000:2018, and FSSAI regulations. Only trained and authorized personnel may operate or work on this equipment.

1.1 Purpose of This Manual

This manual provides complete guidance for safe operation, preventive maintenance, troubleshooting, and emergency procedures for the above equipment. All personnel working on or near this equipment must read and understand this manual before starting work.

2. Technical Specifications

Parameter	Value
PLC	S7-1500
Voltage	415 V
IP Rating	IP54

2.1 Attached Sensors and Instruments

Sensor / Parameter	Monitoring Mode	Data System
Motor Voltage	Continuous monitoring	SCADA / Historian
Power Consumption	Continuous monitoring	SCADA / Historian
Humidity	Continuous monitoring	SCADA / Historian
Machine Temperature	Continuous monitoring	SCADA / Historian

2.2 Design Parameters and Limits

Design Parameter	Value
Minimum Operating Temperature	0°C
Maximum Operating Temperature	153°C
Design Pressure	6 bar(g)
Maximum Allowable Working Pressure (MAWP)	15 bar(g)
Noise Level (at 1m)	67 dB(A)
IP Protection Class	IP67
Hazardous Area Classification	Zone 2 (if applicable)

3. Safety Instructions and Warnings

! DANGER

DANGER: This equipment contains rotating parts, high pressure, and electrical hazards. Failure to follow safety instructions may result in severe injury or death. ALWAYS apply Lockout/Tagout (LOTO) before any maintenance work. See SOP-SAF-002.

! WARNING

FOOD SAFETY: This equipment contacts food-grade product. Only food-grade lubricants, gaskets, and cleaning chemicals approved by FreshFlow QA department may be used. Unauthorized materials may cause product contamination.

3.1 Mandatory PPE

- Safety helmet (bump cap within the production area)
- Safety shoes (steel-toed, anti-slip, food grade)
- Safety glasses / goggles (when working with chemicals or pressurised systems)
- Cut-resistant gloves (when handling sharp parts - NOT when working near rotating machinery)
- Hearing protection (>=85 dB areas)
- High-visibility vest when working in forklift zone

3.2 Lockout / Tagout (LOTO)

1. Notify the shift supervisor and control room before starting any maintenance.
2. Isolate all energy sources: electrical (main isolator), pneumatic (air supply valve), steam (isolation valve), and hydraulic as applicable.
3. Apply personal padlock and danger tag on each isolation point.
4. Verify zero energy state: test electrical with approved voltage tester, release trapped pressure.
5. Perform maintenance work.
6. Remove all tools, materials and personnel from equipment before re-energizing.
7. Remove LOTO in reverse order. Notify operator before restart.

! CAUTION

Never bypass safety interlocks, guards, or alarms for any reason. Bypassing interlocks requires a formal bypass authorization form signed by the Plant Manager.

4. Installation and Commissioning

4.1 Foundation and Leveling

Ensure the concrete foundation is fully cured (minimum 28 days) before placing equipment. Level the baseplate with stainless steel shims. Maximum allowable deviation is 0.05 mm/m. Perform a foundation bolt check to ensure all anchor bolts are correctly embedded and tensioned to the design torque specified in the equipment drawing.

4.2 Piping and Connection Requirements

All piping connected to this equipment must be independently supported to avoid imposing mechanical loads on the equipment nozzles. Maximum allowable nozzle loads must not exceed values specified by the manufacturer. Use food-grade gaskets (EPDM or PTFE) at all flanged joints. Ensure all pipe work is clean and free from debris before connection.

4.3 Electrical Connection

Electrical connections must be performed by a certified electrician in accordance with IS:732, the Indian Electricity Rules, and FreshFlow Beverages Electrical Safety Standards. Verify voltage, frequency, and phase sequence before energizing. Confirm protective earth is correctly connected. Record installation details in the Equipment History Register.

4.4 Pre-Commissioning Checks

1. Verify all mechanical fasteners are tightened to specified torques.
2. Check direction of rotation before full coupling - bump test only.
3. Fill lubricant/oil to correct level per sight glass. Use OEM-specified grade only.
4. Open all isolation valves to the correct position per P&ID.
5. Verify all instrument connections (pressure gauges, temperature sensors, flow meters).
6. Test all alarms and interlocks with SCADA team before commissioning at load.
7. Complete and sign the Pre-Commissioning Checklist (Form FFB-ENG-PC-01).

5. Startup Procedure

5.1 Pre-Start Checks

- Confirm LOTO has been removed and all tools are cleared from equipment.
- Verify no active alarms on CP102 on SCADA / local panel.
- Confirm lubricant and cooling fluid levels are within operating range.
- Confirm Utilities department is ready to receive flow/product.
- Check all guards are fitted securely. Do not start with any guard removed.
- Verify compressed air supply pressure is 6-7 bar (for pneumatic components).

5.2 Startup Sequence

1. Acknowledge all start permissives on SCADA for equipment CP102.
2. Start cooling / utility systems first if applicable (chilled water, steam, air).
3. Engage the main driver at low speed (if VFD) or direct start as per control configuration.
4. Gradually bring equipment to operating setpoint. Monitor all parameters.
5. Verify operating parameters are within normal range within 5 minutes of start.
6. Log startup time in shift log and notify the Utilities supervisor.

! WARNING

If any alarm activates during startup, stop the equipment immediately. Investigate the cause before attempting a second start. Do not repeatedly attempt to force start without investigating.

6. Normal Operating Procedure

Under normal operating conditions, CP102 Control Panel should operate continuously without operator intervention. The SCADA system monitors all critical parameters and will alarm if any value exceeds the set limits. The operator must:

- Perform hourly visual checks of the equipment during operation.
- Verify the running parameters against the normal operating range in Section 2.
- Record any unusual observations in the shift log immediately.
- Report any abnormal noise, vibration, leakage or temperature to the maintenance team.
- Never attempt to adjust process parameters beyond the approved setpoints without authorization.

7. Shutdown Procedure

7.1 Planned Shutdown

1. Notify the Utilities supervisor and control room of the planned shutdown.
2. Reduce equipment to minimum load / speed before stopping.
3. Initiate stop command from SCADA or local panel as appropriate.
4. Allow equipment to coast to rest naturally - do not apply manual braking.
5. Close isolation valves (suction/discharge/utility) as per P&ID.
6. For food contact equipment: Initiate CIP sequence if required before shutdown.
7. Apply LOTO if the shutdown is for maintenance purpose.
8. Log shutdown time, reason, and duration in the equipment history record.

7.2 Emergency Shutdown

! DANGER

Emergency stop buttons are located on the local control panel and at the filling area main panel. Press the red E-STOP button only in case of emergency. E-STOP activates the electrical isolation of all drives simultaneously.

8. Preventive Maintenance Schedule

Task Description	Frequency	Responsible	Record
Visual inspection - leaks, damage, loose fasteners	Daily	Line Operator	Operator Log
Check lubricant levels and colour	Daily	Line Operator	PM Checklist
Vibration and temperature monitoring	Weekly	Reliability Technician	Historian
Clean exterior and ventilation / heat exchange area	Monthly	Maintenance	PM Work Order
Lubricate bearings/chain per OEM schedule	Monthly	Maintenance	PM Work Order
Inspect seals, gaskets and O-rings for wear	Monthly	Maintenance	PM Work Order
Check alignment and coupling condition	Quarterly	Maintenance	PM Work Order
Filter element inspection/replacement	Quarterly	Maintenance	PM Work Order
Oil analysis / oil change	6 Months	Reliability Tech	Oil Sample Report
Foundation bolt torque verification	Yearly	Maintenance	Torque Record
Comprehensive overhaul / OEM inspection	3 Years	OEM Specialist	Overhaul Report

All preventive maintenance must be performed under Lockout/Tagout (SOP-SAF-002) and recorded in the FreshFlow CMMS system with work order number, technician ID, and parts used. Any deviations from the above schedule require Plant Manager approval.

9. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Performance drop	Fouling or wear	Clean or overhaul equipment as per scheduled maintenanc
Unusual noise/vibration	Mechanical looseness or bearing issue	Inspect, tighten fasteners and replace bearings as requ
Overheating	Lack of lubrication or cooling failure	Check lube levels/coolant. Verify cooling flow
Control/sensor failure	Sensor drift or wiring fault	Calibrate or replace sensor. Check wiring continuity
Leakage (product or fluid)	Seal or gasket degradation	Replace gasket/seal. Use food-grade material per plant

If the above corrective actions do not resolve the problem, escalate to the Reliability Engineer (EMP016 - Ajay Mane) or contact the OEM technical support. Record all troubleshooting steps in the CMMS maintenance log.

10. CIP and Sanitation Requirements

All equipment that contacts food product must undergo Clean-In-Place (CIP) at the defined frequency. CIP chemicals, concentrations, temperatures, and contact times are defined in the FreshFlow CIP Master Plan (FFB-QA-CIP-001).

10.1 Standard CIP Parameters

Parameter	Value
Pre-rinse with water	5 min | 25°C | until runoff is clear
Caustic wash (NaOH)	20 min | 75°C | 2% NaOH (food grade)
Intermediate water rinse	5 min | 25°C | until pH neutral
Acid rinse (HNO3)	15 min | 65°C | 1% HNO3 (passivation)
Final water rinse	10 min | 25°C | RO water (conductivity <20 uS/cm)
Sanitisation (Peracetic Acid)	10 min | 25°C | 150 ppm PAA (if required)

! NOTE

CIP verification is mandatory after every CIP cycle. The final rinse water must test <=20 uS/cm conductivity AND pH 6.5-7.5 AND a microbiological swab must confirm <10 CFU/cm2 before the equipment is released for production.

11. Spare Parts Catalog

Part Description	Quantity (Min Stock)	OEM Notes
Mechanical Seal / Seal Cartridge	1 set	Specific to FF-CP-244
Bearing (Drive End)	1 no.	Per OEM specification
Bearing (Non-Drive End)	1 no.	Per OEM specification
Coupling Spider / Insert	1 no.	Per coupling model
Gasket Set	1 set	Full set per overhaul schedule
O-Ring Kit	1 set	Food-grade material only
Oil Filter Element	2 nos.	Per OEM part number
Air Filter Element	2 nos.	Per OEM part number
Pressure Gauge (0-10 bar)	1 no.	Class 1.0 accuracy
Thermocouple / RTD	1 no.	Match to existing sensor type

All spare parts must be sourced from authorized suppliers and must be food-grade compatible where applicable. Store spares in clean, dry conditions in the FreshFlow Spare Parts Store (Block K). Minimum stock levels must be maintained at all times.

12. Alarms, Interlocks and Set Points

Alarm / Interlock	Set Point	Required Action
Temperature High (Warning)	78°C	SCADA alarm - investigate
Temperature High-High (Trip)	104°C	Auto-trip - investigate before restart
Pressure High	12 bar	SCADA alarm - check relief valve
Vibration High (Warning)	7 mm/s	Investigate bearing/alignment
Vibration High-High (Trip)	11 mm/s	Auto-trip - do not restart until investigated
Motor Current High	110% FLC	SCADA alarm - check mechanical load
Low Pressure	3 bar	Check supply / suction conditions
Leak Detected	Float switch / flow deviation	Stop equipment - investigate immediately

13. Emergency Procedures

13.1 Product Spill / Contamination

In the event of a product leak or contamination event: Stop the filling/transfer immediately. Isolate and hold all product affected. Notify QC Manager (EMP003). Do not release any product until QC clearance is obtained. Record in incident log.

13.2 Chemical Spill (Caustic / Acid)

In case of CIP chemical spill: Alert personnel in the area immediately. Evacuate if large volume (>10 litres). Do not enter without chemical-resistant PPE (nitrile gloves, face shield, apron). Neutralize caustic with dilute acid and vice versa. Wash area with large volumes of water. Notify HSE Officer (EMP009 / EMP017).

13.3 CO2 Gas Leak (if applicable)

If CO2 detector GD101 alarms: Evacuate the CO2 zone immediately. Do not re-enter until GD101 reading below 0.5% and confirmed by HSE Officer. CO2 is heavier than air and accumulates at floor level - it causes asphyxiation. Activate exhaust fan EX101 remotely from Control Room. See SOP-SAF-001.

14. Appendix - Reference Documents

- P&ID Drawing: FFB-ENG-PID-CP102-001 Rev 3
- Electrical Drawing: FFB-ENG-EL-CP102-001 Rev 2
- Foundation Drawing: FFB-ENG-CV-CP102-001 Rev 1
- OEM Manual: Siemens FF-CP-244 Original OEM Manual
- CMMS Tag: CP102 - FreshFlow Beverages CMMS
- Risk Assessment: FFB-HSE-RA-CP102-001
- Calibration Record: FFB-QA-CAL-CP102-001
- CIP Validation: FFB-QA-CIP-CP102-001

For the most current version of this manual and related documents, access the FreshFlow Beverages Document Management System (DMS) at the Control Room terminal or contact the Engineering Department.

A. Energy Efficiency Notes

The CP102 Control Panel consumes approximately 22.0 kW at full load. Energy efficiency improvements can be achieved through VFD speed optimization, correct operating point selection, and regular maintenance to minimize friction losses. Track kWh/hour in the SCADA historian and compare with baseline established at commissioning.

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A. Food Safety Considerations

All equipment in product contact zones must be maintained to prevent contamination risks. This includes: using only food-grade lubricants (NSF H1 certified), ensuring all seals and gaskets are food-grade materials (EPDM, PTFE or silicone), maintaining equipment hygiene zones as per the FreshFlow Hygienic Equipment Standard (HES-001), and reporting any metallic part damage immediately to QC to trigger a metal contamination investigation protocol.

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A. Environmental and Sustainability Notes

FreshFlow Beverages is committed to environmental sustainability. All maintenance activities must minimize waste generation. Used oil, chemicals and filters must be disposed of through the approved hazardous waste contractor (GreenEco Pvt. Ltd.) per the Waste Disposal Plan (FFB-HSE-WDP-001). Water usage during CIP must be tracked and reported monthly.

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A. Revision History

Rev 1.0 (2021-10-01): Initial release at plant commissioning.
Rev 1.5 (2022-06-01): Updated troubleshooting section after first year operational review.
Rev 2.0 (2023-07-01): Major update - added CIP parameters and food safety section.
Rev 2.1 (2024-01-01): Alarm setpoints updated per Reliability Engineering recommendation.

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A. Technical Notes - Alignment Standards

This section provides additional technical guidance for maintenance engineers. Vibration analysis using route-based data collection (monthly) and online monitoring (continuous) allows early detection of bearing defects, misalignment, imbalance, and looseness. ISO 10816-3 provides the vibration acceptance criteria for industrial machinery. For CP102 Control Panel, the acceptable vibration limit is 4.5 mm/s RMS at the bearing housing. Any reading above 7.1 mm/s requires immediate investigation and remedial action. Lubrication selection must follow the OEM specification. For rolling element bearings, use lithium complex grease NLGI Grade 2 with food-grade additive package (NSF H1). The regreasing interval is calculated based on the Lubrication Engineer's formula incorporating speed, bearing size, and operating temperature.

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